Fields and Waves

Exam 1 Solutions

Spring 2026

1	**/8**
2	**/8**
3	**/8**
4	**/8**
Total	**/32**

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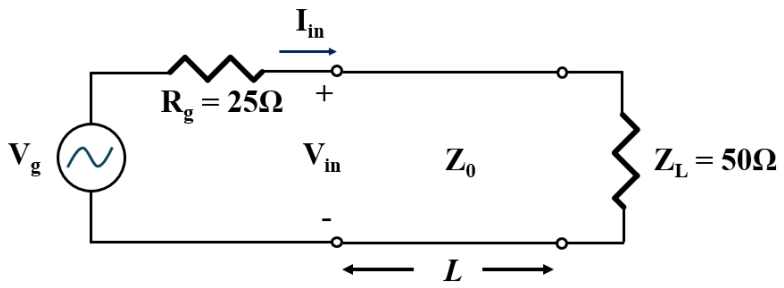
Notes:
SHOW ALL WORK. BEGIN WITH FORMULAS, THEN SUBSTITUTE VALUES AND UNITS. No credit will be given for numbers that appear without justification. For questions that request it, write down the primary equation from the crib sheet that you will use to solve that part of the problem. Partial credit will be awarded where possible, so show your work so that we can follow your thought process. Use the backs of pages if there is not enough room on the front. If you do, please indicate on the front of your exam that there is more work on the back of pages.

Unless otherwise stated in a problem, provide at least 2 significant digits in answers. Ensure that your calculator is set to use radians for trigonometric functions (and not degrees) and include units with all numerical answers.

At least skim through the entire exam before you begin and then start with the problems you know best.

Problem 1: Low-Loss Transmission Line Model [8 pts]

You are given a spool of coaxial cable of length $L = 100\text{m}$ and are tasked with determining its properties through measurements. A diagram of your measurement setup is shown below. Assume the coaxial cable can be treated as a *low-loss transmission line*.



a) [1 pt] You measure the time delay of your cable to be $T_d = 392\text{ns}$. What is the velocity factor of the coaxial cable?

$$u = \frac{L}{T_d} = 2.55 \times 10^8 \frac{\text{m}}{\text{s}} = \mathbf{0.85}c$$

velocity factor =	$\mathbf{0.85}$

b) [1 pt] You measure the amplitude of the voltage at the input to the coaxial cable V_{in} to be 1V for any frequency you set for V_g. What is the characteristic impedance Z_0 of the cable?

$Z_0 =$	$\mathbf{50\Omega}$

Since the amplitude of the voltage does not change with frequency, there is no reflection, indicating that $Z_0 = Z_L = 50\Omega$

c) [1 pt] Calculate the wavelength λ of the voltage wave on the coaxial cable if the frequency of the generator is $f = 100\text{MHz} = 100 \times 10^6$ Hz.

Primary Equation	$\boldsymbol{\lambda = u/f}$

$$\lambda = \frac{u}{f} = \frac{2.55 \times 10^8 \frac{\text{m}}{\text{s}}}{100 \times 10^6\ \text{Hz}} = \mathbf{2.55m}$$

$\lambda =$	$\mathbf{2.55m}$

d) [2 pts] Calculate the per-unit-inductance l and per-unit-capacitance c of the coaxial cable.

Primary Equations	$Z_0 = \sqrt{\frac{l}{c}}$ and $u = \frac{1}{\sqrt{lc}}$

$c = \frac{1}{Z_0 u}$ and $l = \frac{Z_0}{u}$

$c = \frac{1}{50\Omega \cdot 0.85c} = \mathbf{78.4\ pF}$ and $l = \frac{Z_0}{u} = \mathbf{196.1\ nH}$

$l =$	$\mathbf{196.1 nH/m}$

$c =$	$\mathbf{78.4 pF/m}$

e) [2 pts] If you measure the amplitude of the voltage at the load to be $V(L = 100\text{m}) = 0.904\text{V}$ and the amplitude of the voltage at the input to be $V(L = 0\text{m}) = 1\text{V}$, determine the attenuation coefficient α.

$$V(z) = V(z = 0)e^{-\alpha z} \rightarrow V(z = 100\text{m}) = 0.904V = 1Ve^{-\alpha 100\text{m}}$$

$$\alpha = -\frac{\ln(0.904)}{100\text{m}} = \mathbf{0.001 m^{-1}}$$

$\alpha =$	$\mathbf{0.001 m^{-1}}$

f) [1 pt] Calculate the per-unit-length resistance r of the coaxial cable.

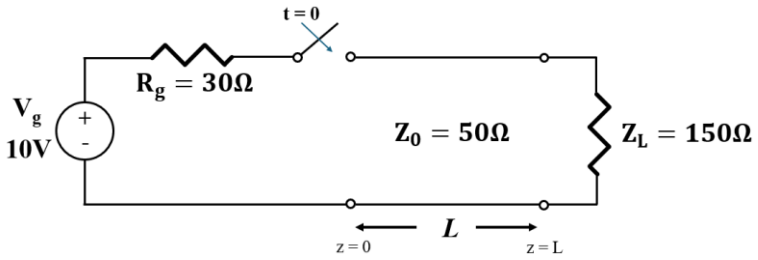
Primary Equation	$\alpha = \frac{r}{2Z_0}$

$$r = 2\alpha Z_0 = 2 * 0.001\text{m}^{-1} * 50\Omega = \mathbf{0.1}\,\frac{\boldsymbol{\Omega}}{\boldsymbol{m}}$$

$r = \mathbf{0.1}\,\frac{\boldsymbol{\Omega}}{\boldsymbol{m}}$

Problem 2: Transient Analysis [8 pts]

A transmission line of characteristic impedance $Z_0 = 50\Omega$ and time delay $T_d = 50\text{ps}$ is connected between a generator circuit with voltage $V_g = 10\text{V}$ and generator impedance $R_g = 30\Omega$, and a load impedance $Z_L = 150\Omega$, as shown below. The switch closes at time $t = 0$ and provides a DC voltage to the transmission line.



a) [2 pts] Calculate the following quantities:

- The reflection coefficient at the load Γ_L
- The reflection coefficient at the generator Γ_g

Primary Equation	$\Gamma = \dfrac{Z - Z_0}{Z + Z_0}$

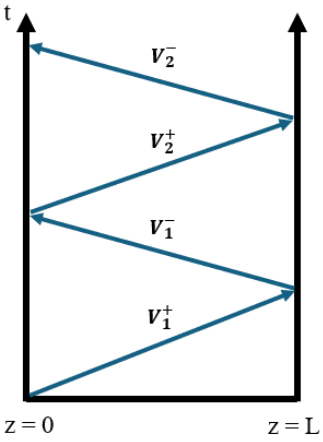
$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{150\Omega - 50\Omega}{150\Omega + 50\Omega} = \mathbf{0.5}$$

$$\Gamma_g = \frac{Z_g - Z_0}{Z_g + Z_0} = \frac{30\Omega - 50\Omega}{30\Omega + 50\Omega} = \mathbf{-0.25}$$

$\Gamma_L = \mathbf{0.5}$

$\Gamma_g = \mathbf{-0.25}$

b) [2 pts] Calculate numerical values for V_1^+, V_1^-, V_2^+ and V_2^- in the bounce diagram below.



$$V_1^+ = V_g \frac{Z_0}{R_g + Z_0} = 10V \frac{50\Omega}{30\Omega + 50\Omega} = \mathbf{6.25V}$$

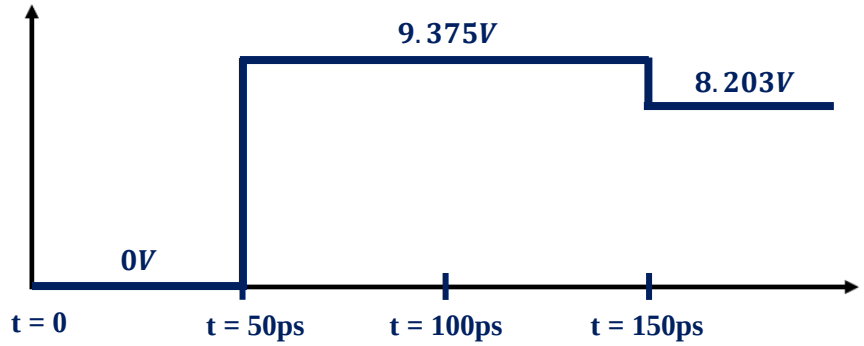
$$V_1^- = \Gamma_L V_1^+ = 0.5 \cdot 6.25V = \mathbf{3.125V}$$

$$V_2^+ = \Gamma_g V_1^- = -0.25 \cdot 3.125V = \mathbf{-0.781V}$$

$$V_2^- = \Gamma_L V_2^+ = 0.5 \cdot (-0.781V) = \mathbf{-0.391V}$$

$V_1^+ = \mathbf{6.25V}$
$V_1^- = \mathbf{3.125V}$
$V_2^+ = \mathbf{-0.781V}$
$V_2^- = \mathbf{-0.391V}$

c) [3 pts] Sketch a plot of the voltage at the load $V(z = L)$ vs. time for $0 \leq t \leq 150\text{ps}$. Label voltage amplitudes and critical times, such as when the voltage wave undergoes relevant reflections.



$V_L(t = 50\text{ps}) = V_1^+ + V_1^- = 6.25V + 3.125V = \mathbf{9.375V}$

$V_L(t = 150\text{ps}) = V_1^+ + V_1^- + V_2^+ + V_2^- = 9.375V - 1.172V = \mathbf{8.203V}$

d) [1 pt] Calculate the voltage on the transmission line after it has reached steady state V_{ss}.

Primary Equation $V_\infty = V_g \dfrac{Z_L}{R_g + Z_L}$

$V_\infty = 10V \dfrac{150\Omega}{30\Omega + 150\Omega} = \mathbf{8.33V}$

$V_{ss} = \mathbf{8.33V}$

Problem 3: Input Impedance [8 pts]

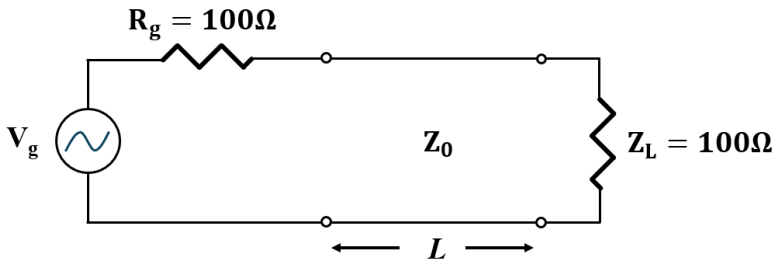
For all circuits in this problem, the frequency of the generator voltage is $f = 200\text{MHz} = 200 \times 10^6\text{Hz}$ and the velocity factor of all coaxial cables is 0.65. In each part, provide a numerical answer with units.

a) For part a) only: you have access to spools of coaxial cable with characteristic impedances:

- Cable type #1: $Z_{01} = 25\Omega$
- Cable type #2: $Z_{02} = 50\Omega$
- Cable type #3: $Z_{03} = 75\Omega$

In each of the circuits below, specify the type of cable (which characteristic impedance Z_0) and length L that match the generator impedance to the load impedance.

i) [2 pts] Circuit #1



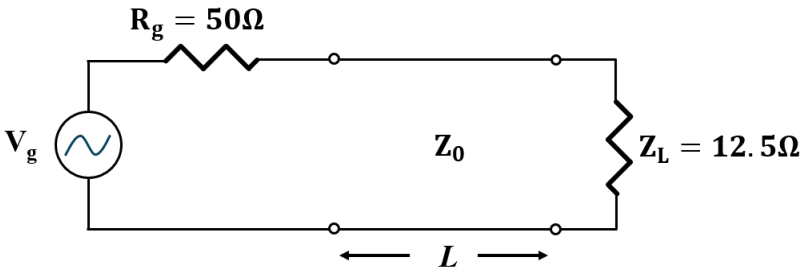
$Z_0 = \mathbf{25\Omega, 50\Omega, 75\Omega}$
$L = \mathbf{0.4875m}$

Since a line of length $\lambda/2$ has an input impedance equal to Z_L, any coaxial cable type will lead to a match.

$$\lambda = \frac{u}{f} = \frac{0.65 \cdot 3 \times 10^8 \frac{\text{m}}{\text{s}}}{200 \times 10^6\ \text{Hz}} = 0.975\text{m}$$

$$L = \lambda/2 = \mathbf{0.4875m}$$

ii) [2 pts] Circuit #2



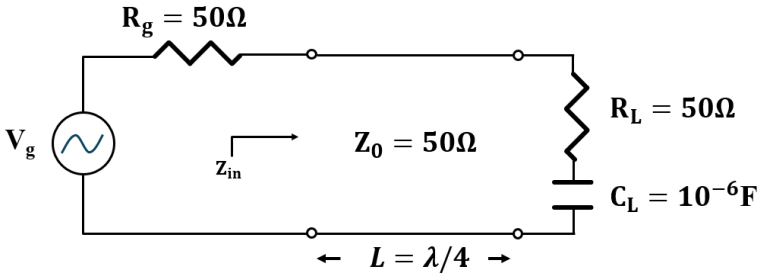
$Z_0 = \mathbf{25\Omega}$
$L = \mathbf{0.2438m}$

A quarter wave transformer $L = \lambda/4$ will match two resistive loads.

$$Z_{in}^{\lambda/4} = \frac{Z_0^2}{Z_L} \rightarrow Z_0 = \sqrt{Z_L Z_{in}^{\lambda/4}} = \sqrt{12.5\Omega \cdot 50\Omega} = \mathbf{25\Omega}$$

$$L = \lambda/4 = \mathbf{0.2438m}$$

b) [2 pts] Determine Z_{in} for the circuit below.



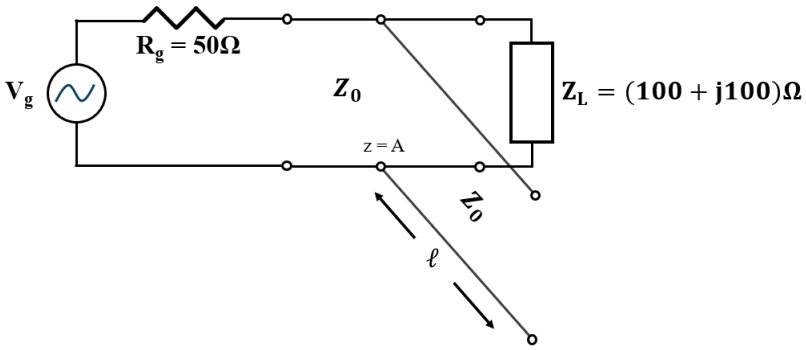
Primary Equation	$Z_{in}^{\lambda/4} = \dfrac{Z_0^2}{Z_L}$

$$Z_L = 50\Omega + 1/(j2\pi \cdot 200 \times 10^6 \text{Hz} \cdot 10^{-6}\text{F}) = (50 - \text{j}7.96 \times 10^{-4})\Omega$$

$$Z_{in}^{\lambda/4} = \frac{Z_0^2}{Z_L} = \frac{(50\Omega)^2}{(50 - \text{j}7.96 \times 10^{-4})\Omega} = \mathbf{(50 + j7.96 \times 10^{-4})\Omega}$$

$$\boxed{Z_{in} = \mathbf{(50 + j7.96 \times 10^{-4})\Omega}}$$

c) [2 pts] In the circuit below, what length of open-circuited cable ℓ would lead to full reflection of the voltage wave at $z = A$?



Primary Equation	$Z_{in}^{\lambda/4} = \dfrac{Z_0^2}{Z_L}$

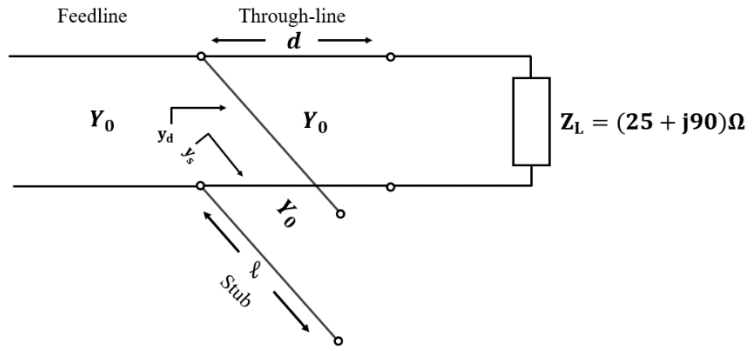
A quarter wave transformer $L = \lambda/4$ will transform an open circuit load to look like a short circuit impedance at the input, which will lead to a reflection coefficient of $\Gamma = -1$, so

$$L = \lambda/4 = \mathbf{0.2438m}$$

$$\boxed{\ell = \mathbf{0.2438m}}$$

Problem 4: Single-Stub Matching with a Smith Chart [8 pts]

Use the included Smith chart to match the feedline's characteristic impedance Z_0 to the load impedance Z_L via the single-stub matching method. As shown below, the matching circuit consists of a through-line of length d, which is connected to the load and an open-circuited stub of length ℓ, connected in parallel with the through-line. $Z_0 = 50\Omega$ for all transmission lines in this problem.



a) [2 pts] Calculate the normalized load impedance z_L and plot it on the Smith chart.

Primary Equation $\boldsymbol{z_L = \frac{Z_L}{Z_0}}$

$$z_L = \frac{Z_L}{Z_0} = \frac{(25 + j90)\Omega}{50\Omega} = \mathbf{0.5 + j1.8}$$

(See Smith chart for location of z_L)

$\boldsymbol{z_L = 0.5 + j1.8}$

b) [1 pt] Determine and plot the location of the normalized load admittance y_L on the Smith chart, then read off its numerical value.

$\boldsymbol{y_L = 0.14 - j.52}$

(See Smith chart for location of y_L)

c) [3 pts] Determine the length of the through-line d that yields an input admittance whose real part is matched to the characteristic admittance $Y_0 = 1/Z_0$ (i.e. $y_d = 1 + jA$). Show all work for how you determined this length on the Smith chart. You may leave your answer in terms of wavelengths.

$$d = (0.5\lambda - 0.423\lambda) + 0.203\lambda = \mathbf{0.271\lambda}$$

(See Smith chart for y_d)

$\boldsymbol{d = 0.271\lambda}$

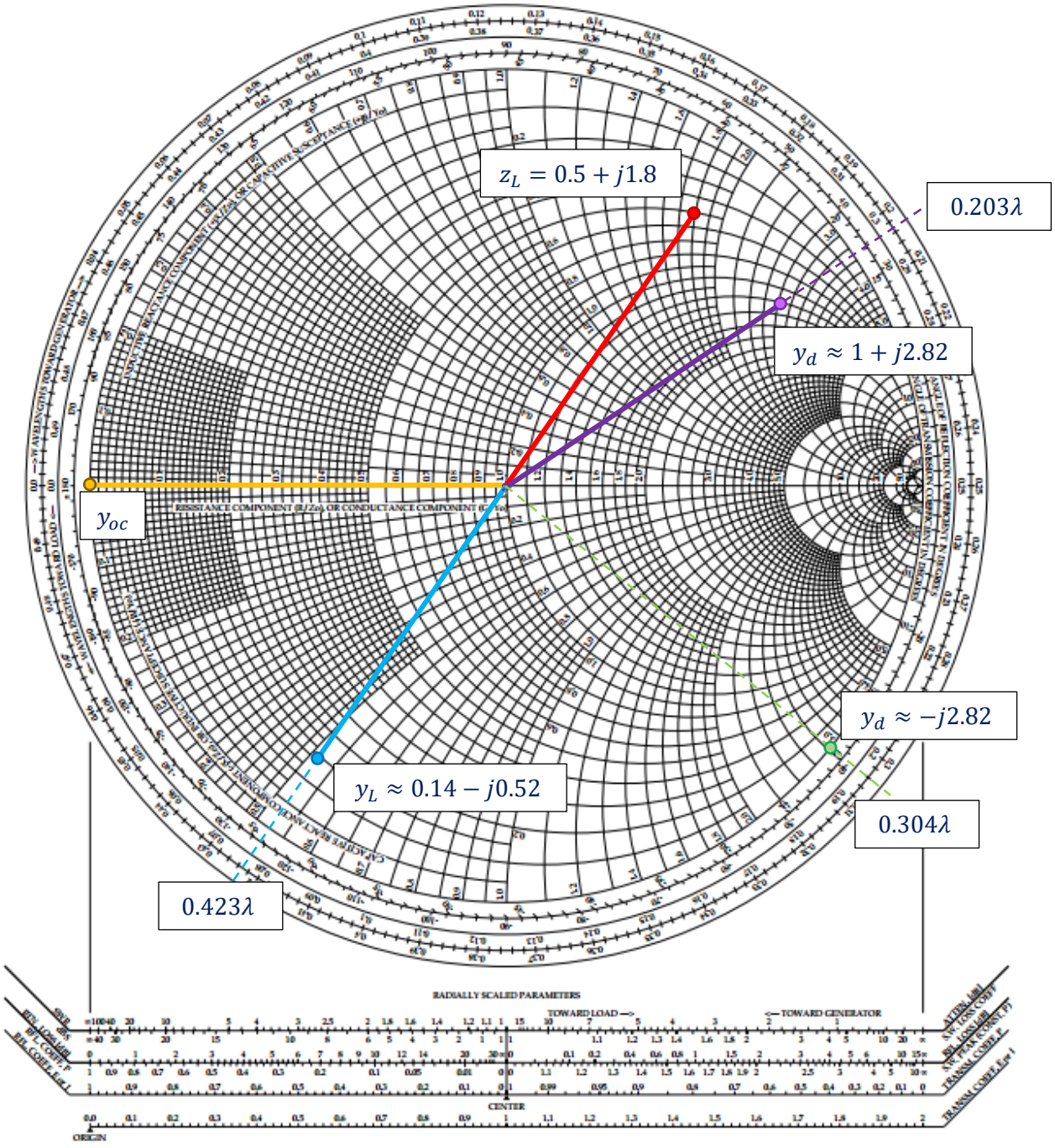
Alternate solution: $y_d = 1 - j2.82$ and $d = 0.318\lambda$

d) [2 pts] Determine the length of the open-circuited stub ℓ that completes the impedance match. Show all work for how you determined this length on the Smith chart. You may leave your answer in terms of wavelengths.

(See Smith chart for y_ℓ)

$\boldsymbol{\ell = 0.305\lambda}$

Alternate solution for $y_d = 1 - j2.82$: $\ell = 0.196\lambda$


The Complete Smith Chart
Black Magic Design
$z_L = 0.5 + j1.8$
0.203λ
$y_d \approx 1 + j2.82$
y_{oc}
$y_d \approx -j2.82$
$y_L \approx 0.14 - j0.52$
0.304λ
0.423λ
RADIALLY SCALED PARAMETERS
TOWARD LOAD
TOWARD GENERATOR
CENTER
ORIGIN